Building NeXTGen **Responsible AI & Quantum Leaders**

Empowering the Next Generation
with AI-Ready Programs and
Future-Forward Skills

The World Is Changing, So Must **the Way We Learn!**

The world is changing rapidly from self-driving cars to AI-generated content, from algorithms that diagnose diseases to smart assistants simplifying our everyday lives.

This transformation brings exciting opportunities for the next generation. To truly prepare them, we must nurture:

- Strong math foundations
- Creative, logic-based coding skills
- Deep understanding of AI and hands-on possibilities

This shift isn't just important, it's essential. And the earlier we begin, the stronger their future becomes.

Here’s How, **We Help You Grow**

{igebra.ai} is a US-based AI research, development, and education company founded by Srini Vemula—an educator and technologist based in the United States.

He is joined by a team of co-directors and a growing network of 50+ global partners. Together, they bring deep expertise in education, technology, and innovation, delivering the knowledge, tools, and passion needed to help students become truly AI-ready.

Coding4AI

Code Smart. Think AI. Build the Future.

Program Details:

- ☑ Live Online Mini Group Classes
- ☑ Highly Qualified, Passionate & Trainers
- ☑ High Quality Video Backup to Revise Anytime

Group	Grades	Levels	Frequency	Duration	Total Classes
Elementary	Grades 4–5	Level 1, 2, 3	1 Class/Week	90 Minutes	20 Classes
Middle School	Grades 6–8	Level 1, 2, 3	1 Class/Week	90 Minutes	20 Classes
High School	Grades 9–10	Level 1, 2, 3	1 Class/Week	90 Minutes	20 Classes

$499 Per Level

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Topics for Elementary

Level 1

- What Is Python? Writing Your First Line of Code
- Print, Input & Variables: Making the Computer Talk
- Working with Numbers and Strings
- Making Decisions: If-Else Statements
- Repeating Actions: Loops (For and While)
- Drawing with Code: Turtle Graphics
- Using Lists to Store Data
- Fun with Random
- Project

Level 2

- Review of Python Basics
- Error Debugging
- Building Interactive Stories with User Choices
- Drawing Patterns and Shapes with Turtle
- Timers, Counters, and Mini Game Logic
- Creating Simple Animations with Loops
- Project

Level 3

- What Is AI? Real-Life Examples Explained
- Rule-Based AI: Create a Basic Chatbot
- Collecting and Using Data in Python (Lists & Conditions)
- Image Classification with Teachable Machine (Used via Code)
- Voice Commands Using Speech Recognition (Demo-Based)
- Using Randomness to Simulate "Smart" Behavior
- Project

Topics for Middle School

Level 1

- Introduction to Programming & Python
- Variables, Data Types & Inputs
- Operators and Expressions
- Conditionals (if-else, elif)
- Loops (for, while)
- Functions and Reusability
- Lists & Basic List Operations
- Dictionaries and Tuples
- Intro to Strings and String Manipulation
- Error Handling & Debugging
- Creative Coding with Turtle
- File Input/Output
- Simple Game Logic
- Working with Random & Time Modules
- Math with Python (Basic Logic & Arithmetic Apps)
- Introduction to IDEs (Thonny, Replit, Jupyter)
- Projects

Level 2

- Python Refresher: Review of Level 1
- Advanced Lists & List Comprehensions
- Working with CSV & JSON files
- Using Libraries: NumPy Basics
- Data Visualization with Matplotlib
- DataFrames with pandas
- Statistics Essentials
- Data Cleaning
- Intro to Data Projects: Reading, Cleaning, Plotting
- Intro to Machine Learning
- Build Your First ML Model (Scikit-learn)
- Classification: Predicting Labels (e.g., Iris dataset)
- Regression: Predicting Numbers
- Model Evaluation (Accuracy, Train/Test split)
- Projects

Level 3

- AI vs ML vs DL – Real Examples for Kids
- Neural Networks Simplified (with visuals & games)
- How Image, Text, and Sound Work in AI
- Intro to TensorFlow/Keras (Simplified API)
- Build a Simple Neural Network
- Train on Image Data (Fashion MNIST)
- Train on Text Data (Movie Reviews or Emoji Text)
- Audio Projects: Recognize Sound or Voice
- Intro to Generative AI and Create Content
- Chatbots with pre-trained models (using Hugging Face or OpenAI API if allowed)
- Projects
- Build a Portfolio Website

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Topics for High School

Level 1

- Python Setup, Syntax, and IDEs
- Variables, Data Types, Type Conversion
- Operators and Logical Expressions
- Control Flow: Conditionals (if-elif-else)
- Loops (for, while) and Nested Loops
- Functions with Parameters and Return Values
- String Operations and Formatting
- Lists, Indexing, and List Comprehensions
- Dictionaries, Sets, and Tuples
- Modules, Importing Libraries, Random, Time
- Error Handling with try/except
- File Input/Output
- Object-Oriented Programming Basics (Classes and Objects)
- Intro to Recursion and Lambda Functions
- Coding with Turtle Graphics: Creative Design
- Simple Terminal-Based Games (Hangman, Tic Tac Toe)
- Building with External APIs (Weather, Jokes, Quotes)
- Project Planning and Debugging Strategies
- Build Your Own Command-Line Tool
- Capstone Project

Level 2

- Recap: Python Structures & OOP Principles
- Working with Data (CSV, JSON, APIs)
- NumPy: Arrays, Broadcasting, Math Operations
- pandas: DataFrames, Indexing, GroupBy, Filtering
- Data Cleaning & Wrangling
- Exploratory Data Analysis: Summary Stats, Correlations
- Matplotlib and Seaborn for Visualization
- Intro to Probability and Statistics for AI
- Intro to Machine Learning Concepts
- scikit-learn Overview
- Supervised Learning: Classification (KNN, Decision Trees)
- Supervised Learning: Regression (Linear, Polynomial)
- Unsupervised Learning: Clustering (KMeans)
- Model Evaluation Metrics: Accuracy, Precision, Recall
- Train/Test Split, Cross-Validation
- Feature Engineering & Selection
- Build ML App with Streamlit/Gradio
- Build a Recommendation System
- Real-World Dataset Exploration (e.g., COVID, Climate, Finance)
- Intro to AI Ethics, Fairness, Bias, Transparency
- Capstone Project

Level 3

- Recap: Machine Learning Pipeline & Ethics
- Intro to Neural Networks & Activation Functions
- Deep Learning with TensorFlow/Keras
- Build & Train a Fully Connected Network
- Overfitting, Regularization, and Dropout
- Convolutional Neural Networks (CNNs) for Image Classification
- Image Project: Recognize Handwritten Digits or Faces
- Recurrent Neural Networks (RNNs) and Text Prediction
- Transformers and Language Models (Intro to GPT/BERT)
- Generative AI Overview (Text, Image, Audio)
- Prompt Engineering Basics
- Text Generation Project (Poetry, Story, Chatbot)
- Image Generation with DALL·E/Stable Diffusion
- Fine-Tuning Pre-trained Models
- Build a Voice Assistant or AI Podcast Narrator
- Ethical Challenges in Generative AI (Deepfakes, Misinformation)
- AI for Good: Projects with Social Impact
- Use of AI in Careers: Health, Art, Science, Law
- Capstone Project

Contact Us

Get in Touch

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